

ADITYA PRATAP SINGH TAKULI

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Profile Summary

Software Engineer with hands-on experience building backend systems, REST APIs, and data-driven models in Python, C++, and SQL across relational (PostgreSQL) and NoSQL databases. Strong foundation in Data Structures & Algorithms, OOP, DBMS, and Operating Systems, with a proven record of shipping tested, production-grade software and translating data and performance metrics into measurable business outcomes. Experienced in agile delivery, CI/CD, and collaborative open-source contribution, with additional exposure to AI/ML, agentic AI systems, and workflow automation, and curious about applying engineering and data-led thinking to markets technology.

Education

Vellore Institute of Technology, Chennai

Aug 2023 – Jul 2027 (Expected)

B.Tech – Electronics and Communication Engineering | CGPA: 8.34 / 10 | No active backlogs

St. Theresa Sr. Sec. School (CBSE) — Class 12: 85% | Class 10: 89.4%

Technical Skills

Languages: Python (IIT Bombay Certified), C++ (IIT Bombay Certified), C (IIT Bombay Certified), SQL, Bash/Shell, Git

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Networking Fundamentals, Distributed Systems, Multithreaded Programming

Data & Databases: PostgreSQL, SQLite, MongoDB, Pandas, NumPy, Exploratory Data Analysis, Statistical & Predictive Modelling (XGBoost, Regression)

Backend & Platforms: FastAPI, REST & WebSocket APIs, Docker, CI/CD (GitHub Actions), AWS, Linux

AI/ML & Automation: PyTorch, Scikit-learn, YOLO, RAG, LangChain, LangGraph, Agentic AI Systems, Prompt Engineering

Professional Experience

Software Engineer Intern — *Staunch Technologies Pvt. Ltd.*

Jan 2026 – Present

- Designed and trained detection models on embedded datasets; built scalable data pipelines for dataset generation, preprocessing, and class-imbalance mitigation, improving model robustness (mAP@50: 85%).
- Delivered a production-ready, hardware-software integrated embedded system in collaboration with an enterprise banking partner (Indian Bank), from data pipeline through deployment.

Samsung PRISM Project Intern — *Samsung R&D*

June 2025 – Nov 2025

- Built a multi-output XGBoost predictive model forecasting nine microservice performance metrics (latency, throughput, CPU, memory) across 3,042 samples, achieving $R^2 \approx 0.77$.
- Quantified a $\sim 3.9\times$ throughput gain (17,522 vs 4,487 req/s) under high-concurrency load and converted the benchmarks into business-facing architecture recommendations across 10+ microservices.

Technical Product Lead — *Checkinly (Smart Hospitality Platform)*

June 2025 – Feb 2026

- Led technical development of a check-in platform end-to-end; integrated Onfido identity verification and Tuya/TTLock IoT SDKs, automating identity and access workflows and cutting manual check-in time by 85–90%.
- Ran market and feasibility research (surveys, competitor benchmarking) that directly shaped product architecture and go-to-market decisions.

HPC / Research Intern — *CSIR Fourth Paradigm Institute (CSIR-4PI), Bengaluru*

May 2025 – Aug 2025

- Designed a hybrid HTTP (REST) + WebSocket network using FastAPI for real-time WLAN data synchronisation, achieving sub-second cross-node propagation.
- Optimised deep learning models for deployment on resource-constrained edge hardware, improving inference efficiency and deployment readiness.

Projects & Open Source

LAMDA Analytics — Distributed Supply Chain Analytics | Python | Temporal Graph Networks | Multi-Agent Systems

- Built a real-time analytics system using Temporal Graph Networks to forecast supply-chain disruptions with multi-horizon predictions, incorporating market signals, geopolitical risk, and port data.
- Developed a multi-agent architecture for autonomous data scraping, normalisation, and analysis of real-world supply/demand factors; won the NOKIA CERONIX National Hackathon (top 5% of 878 teams).

HSF/prmon — CERN Open-Source Contribution (Merged PR #302) | C++ | CLI Tooling | Code Review

- Replaced the legacy CLI parser in a CERN/HSF C++ process-monitoring tool used in high-energy physics workflows, removing ~ 60 lines of fragmented code while preserving all 12 CLI options and passing all 14 existing tests.
- Passed two rounds of formal review from a CERN maintainer and was added as an official project contributor.

Gold Detection & Segmentation System | Python | YOLO | OpenCV | SQLite | OCR

- Built a real-time detection pipeline combining a custom YOLO model with a person-segmentation model to suppress false positives, plus OCR-based weight extraction from a live scale feed.
- Designed a SQLite-backed logging layer with deduplication and a companion desktop GUI viewer for browsing detection history, validated with an automated test suite.

Achievements & Certifications

- Winner, NOKIA CERONIX National Hackathon (top 5% of 878 teams, distributed multi-agent analytics system); 2nd Place, CodeInit Hackathon (ACM); represented VIT Chennai at UK Innovate '25.
- Official Contributor, CERN HSF/prmon; Research Paper published (DOI: 10.1134/S1063782625604522); Indian Patent filed (App. No. 202641053086).
- Certified: Python, C, C++ (IIT Bombay); Generative AI Applications (IBM) — Prompt Engineering, LangChain Agents & Tools.